

Abbas Zohrevand

Ph.D. Candidate in Artificial Intelligence

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Research Summary

I am a Ph.D. candidate in Artificial Intelligence at the University of Tehran, under the supervision of Dr. Reshad Hosseini and Prof. Hadi Moradi. Over the course of my research, I have worked across several areas of machine learning and computer vision, including handwriting and document analysis, biometrics, and remote-sensing image understanding. My current work centers on probabilistic models for sequence(time-series) analysis, and I am additionally drawn to optimization theory and the mathematics of manifolds as tools for building more principled and geometrically grounded learning methods. I bring over eight years of university teaching experience alongside an active publication record in machine learning.

Education

Ph.D. in Computer Engineering (Artificial Intelligence)

Sep 2021 – Present

University of Tehran, Tehran, Iran

- Supervisors: Dr. Reshad Hosseini, Prof. Hadi Moradi
- Cumulative GPA: 3.8 / 4.0
- Focus: Machine Learning, Computer Vision, Time Series Analysis

M.Sc. in Computer Engineering (Artificial Intelligence)

Sep 2011 – Feb 2014

Shahrood University of Technology, Shahrood, Semnan, Iran

- Thesis: *Recognition and Localization of Objects in Complex Scenes*
- Supervisor: Dr. Alireza Ahmadyfard
- Cumulative GPA: 3.7 / 4.0

B.Sc. in Computer Engineering

Sep 2006 – Feb 2011

University of Birjand, Birjand, Southern Khorasan, Iran

- Thesis: *Persian Handwritten Digit Recognition Using Gradient Features*
- Supervisor: Dr. Javad Sadri
- Cumulative GPA: 3.5 / 4.0

Publications

Journal Articles

- [J1] A. Zohrevand, J. Sadri, and Z. Imani, "A comprehensive Persian offline handwritten database for investigating the effects of heritability and family relationships on handwriting," *Int. J. Pattern Recognit. Artif. Intell.*, JCR-Q4. *Under review*.
- [J2] A. Zohrevand, Z. Imani, J. Sadri, and C. Y. Suen, "Does color modality affect handwriting recognition? An empirical study on Persian handwritings using convolutional neural networks," *Int. J. Wavelets Multiresolution Inf. Process.*, arXiv:2307.12150.
- [J3] A. Zohrevand and Z. Imani, "Holistic Persian handwritten word recognition using convolutional neural network," *Int. J. Eng.*, JCR-Q2, vol. 34, no. 8, pp. 2028–2037, Aug. 2021.
- [J4] A. Zohrevand, Z. Imani, and M. Ezoji, "Deep convolutional neural network for finger-knuckle-print recognition," *Int. J. Eng.*, JCR-Q2, vol. 34, no. 7, pp. 1684–1693, Jul. 2021.
- [J5] A. Zohrevand, "Probabilistic relaxation labeling: A short survey on object recognition," *Int. J. Comput. Appl.*, vol. 181, no. 38, pp. 40–44, Jan. 2019.
- [J6] Z. Imani, A. Ahmadyfard, and A. Zohrevand, "Holistic Farsi handwritten word recognition using gradient features," *J. AI Data Min.*, ISC, vol. 4, no. 1, pp. 19–25, Feb. 2016.

Conference Proceedings

- [C1] Y. Riyazi, S. M. Sadjadi, A. Zohrevand, and R. Hosseini, “High-resolution remote sensing image captioning based on structured attention and SAM network,” in *Proc. 32nd Int. Conf. Electr. Eng. (ICEE)*, Ahvaz, Iran, May 2024.
- [C2] A. Zohrevand and Z. Imani, “An empirical study of the performance of different optimizers in deep neural networks,” in *Proc. 12th Iranian and 2nd Int. Conf. Mach. Vis. Image Process.*, Ahvaz, Iran, Feb. 2022.
- [C3] A. Zohrevand, J. Sadri, Z. Imani, and M. R. Yeganezad, “Line segmentation in Persian handwritten documents based on a novel projection histogram method,” in *Proc. 4th Int. Conf. Pattern Recognit. Image Anal.*, Tehran, Iran, Mar. 2019.
- [C4] V. Kiani and A. Zohrevand, “A fuzzy directional median filter for fixed-value impulse noise removal,” in *Proc. 7th Iranian Joint Congr. Fuzzy Intell. Syst.*, Bojnord, Iran, Jan. 2019.
- [C5] A. Zohrevand, A. Ahmadyfard, A. A. Pouyan, and Z. Imani, “A SIFT-based object recognition using contextual information,” in *Proc. Iranian Conf. Intell. Syst.*, Bam, Kerman, Iran, Jan. 2014.
- [C6] Z. Imani, A. Ahmadyfard, M. Alipour, and A. Zohrevand, “Offline handwritten Farsi cursive text recognition using hidden Markov models,” in *Proc. 8th Iranian Conf. Mach. Vis. Image Process.*, Zanjan, Iran, Sep. 2013.
- [C7] Z. Imani, A. Ahmadyfard, and A. Zohrevand, “Introduction to database FARSA: Digital images of handwritten Farsi words,” in *Proc. 11th Iranian Conf. Intell. Syst.*, Tehran, Iran, Mar. 2013.

Books

- [B1] A. Zohrevand, *Introduction to Data Structures* (in Persian), Bojnord University. *Under review*.
- [B2] A. Zohrevand and N. Samadiani, *C++ Programming Made Easy* (in Persian), Tehran Institute of Technology, ISBN 978-600-124-985-3, Jan. 2018.
- [B3] A. Zohrevand and H. Mafakheri, *Learn L^AT_EX Step by Step* (in Persian), Tehran Institute of Technology, ISBN 978-622-218-061-4, Feb. 2018.

Academic Experience

Faculty Member – Instructor

Feb 2017 – Present

Department of Computer Engineering, Kosar University of Bojnord

- Courses taught: Artificial Intelligence, Algorithm Design, Discrete Mathematics, Advanced Programming, Data Structures, Operating Systems, Theory of Computation

Guest Lecturer

Sep 2014 – Present

Department of Computer Engineering, University of Bojnord

- Courses taught: Artificial Intelligence, Algorithm Design, Discrete Mathematics, Advanced Programming, Data Structures

Guest Lecturer

Sep 2014 – Feb 2017

Department of Computer Engineering, Kosar University of Bojnord

- Courses taught: Artificial Intelligence, Algorithm Design, Discrete Mathematics, Advanced Programming, Data Structures

Skills & Languages

Programming	Python, C++, C, Java
Software & Libraries	MATLAB, PyCharm, VS Code, PyTorch
Scientific Writing	L ^A T _E X, X _E L ^A T _E X, Microsoft Office
Languages	Persian (native), English (fluent)

Research Interests

Machine Learning · Computer Vision · Deep Learning · Probabilistic Models · Generative Models

References

Dr. Reshad Hosseini

Assistant Professor, Machine Learning and Robotics
Group

University of Tehran

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Dr. Javad Sadri

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puter Science

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